(CPES 3)

TECHNOPRENEURSHIP

ACTIVITY NO. 7

BUSINESS MODEL

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Submitted to:

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Subject Professor

March 19, 2025

BUSINESS MODEL

Designed By: Casimero Group

Date: March 19, 2025

KEY ACTIVITIES

1. Automated Butterfly Habitat Monitoring and Maintenance
- The system continuously tracks environmental factors such as temperature, humidity, and light conditions to ensure optimal rearing conditions for butterflies. Automated alerts notify users of any required adjustments.

2. AI-Powered Food Quality Analysis and Optimization
- The built-in image-based monitoring system analyzes the quality and freshness of butterfly food sources to ensure they meet optimal nutrition standards, reducing waste and improving growth rates.

3. Data Collection and Reporting for Butterfly Growth and Health
- The system records and processes real-time data on butterfly development, health, and environmental conditions. Users can access reports and analytics to make data-driven decisions.

4. Customer Support and Technical Assistance for Users
- A dedicated support team assists users with troubleshooting, software updates, and optimizing their system for better results. This ensures a seamless user experience and long-term customer satisfaction.

5. Continuous AI Model Training and Improvement
- The AI system learns from collected data and user interactions to enhance its ability to detect food quality issues and optimize butterfly growth conditions over time. Regular software updates ensure improved performance.

6. Marketing and Outreach to Butterfly Farmers, Researchers, and Conservationists
-Promotional efforts include social media marketing, partnerships with butterfly conservation groups, and participation in agricultural and ecological conferences to reach target users and expand the customer base.

7. Supply Chain Management for System Components
- Managing the procurement and distribution of essential system components (sensors, cameras, and automation modules) to ensure uninterrupted availability and smooth business operations.

KEY RESOURCES

Tangible Resources

Hardware Components:
- Microcontrollers (Arduino UNO, Raspberry Pi)
- Wi-Fi Module (ESP32)
- Temperature and Humidity Sensors (DHT11, DHT22)
- Light Sensors (Photodiodes, LDR)
- Relays and MOSFET for actuators control \
- Fans, heaters, and humidifiers
- Acrylic or Polycarbonate for the enclosure
- Power Supply (12V DC)
- Wires and Connectors
- Backup Power Supply lRaspberry Pi Camera
- Enclosure Construction Materials

Intangible Resources

Knowledge and Expertise:
- Knowledge of butterfly species, their life cycles, and optimal rearing conditions.
- Expertise in agricultural engineering, Internet of Things (IoT), artificial intelligence, and embedded systems.
- Knowledge of image processing techniques.
- Understanding of control algorithms.

Skills and Abilities:
- Technical skills to design and develop hardware components.
- Software development abilities for image processing and control systems.
- Research capabilities.
- Design and development skills for the enclosure and user interface.
- Testing and evaluation skills.

Intellectual Property:
- Control algorithms
- Image processing algorithms for food quality and quantity detection.

KEY PARTNERS

- Collaboration with Local Butterfly Centers
- Engagement with Farmers
- Support from Local Government

VALUE PROPOSITIONS

"Elevate your butterfly farming with AI care for smarter, healthier growth."

CUSTOMER RELATIONSHIP

- Long-term partnership with personalized support
- Collaborative and knowledge-sharing relationship
- Supportive and mission-driven relationship
- Engaged and educational relationship
- Community-driven and self-service with support
- Institutional partnership

CUSTOMER SEGMENTS

- Butterfly Breeders & Farmers
- Research Institutions & Universities
- Environmental & Conservation Organizations
- Educational Institutions (Schools & Museums)
- Small-Scale Enthusiasts
- Government & Agricultural Departments

CHANNELS

- **Direct Sales to Farmers** – The system will be directly marketed and sold to butterfly farmers through on-site visits, training programs, and demonstrations.
- **Partnerships with Local Government and Agricultural Organizations** – Collaborations with LGUs, the Department of Agriculture, and butterfly conservation centers to support and distribute the system.
- **Online Sales and E-commerce Platforms** – The product will be available on Shopee, Lazada, and Facebook Marketplace, along with a dedicated website for direct orders.
- **Technology and Farming Expos** – Participation in agricultural trade fairs and technology expos to showcase the system through live demonstrations.
- **Business-to-Business (B2B) Sales** – Selling in bulk to butterfly farming startups, agritech businesses, butterfly-themed resorts, and eco-parks.
- **Academic and Research Collaborations** – Partnering with universities and research institutions to integrate the system into agricultural and biological studies.

COST STRUCTURE

Table 1. Project's Production and Development Cost

Component/Service	Quantity	Unit Cost (PHP)	Total Cost (PHP)
Raspberry Pi 4 Model B	1	6,000	6,000
Raspberry Pi Camera Module V2.1	2	5,000	10,000
ESP32 Wi-Fi Module	1	500	500
DHT11 Temperature and Humidity Sensor	4	50	200
Photodiode	4	50	200
Relays	4	50	200
MOSFETs	4	50	200
Fans	2	200	400
Heaters	2	300	600
Humidifiers	2	500	1,000
Acrylic Sheets (for enclosure)	3 sq. meters	300	900
12V DC Power Supply	1	1,000	1,000
Wires and Connectors	2 sets	200	400
Backup Power Supply	1	5,000	5,000
Miscellaneous Cost			
Enclosure Construction Materials	-	-	3,000
Shipping and Handling Fees	-	-	1,000
Prototype Testing and Iteration	-	-	5,000
Contingency Fund	-	-	5,000
Estimated Total Cost			PHP 40, 600

COST STRUCTURE

Cost Breakdown by Stage

I.Setting Up//Research and Development

Fixed Cost

- Prototype Development Cost: PHP 40, 600.00
- Software Development Tools: PHP 5,000.00

Variable Cost

- Prototype Components: Included in the prototype development cost bat may vary PHP 5, 000.00 – PHP 10, 000.00 per system depending on the component's quality, availability and manufacturer

II.Production/Manufacturing

Fixed Cost:

- Equipment Purchase (Specialized tools for enclosure construction): PHP 8, 000.00
- Website Development/Maintenance: PHP 10,000.00

Variable Cost

- Component Costs (Microcontrollers, Raspberry Pi, Sensors): PHP 26, 200.00 (initial calculation)
- Assembly Labor (if hiring assemblers): Varies per system

III.Marketing and Distribution

Fixed Cost

- Marketing Materials (Brochures, etc.): PHP 3,000.00

Variable Cost

- Sales Commissions: Varies depending on the percentage of sales
- Shipping Cost: Varies per system and location

IV.Operational Costs

Fixed Cost

- Business Permits and Licenses: PHP 1,000.00 but may vary by locality

Variable Cost

- Warranty/Repair Cost: Varies depending on the cost of replacement parts and labor.

REVENUE STREAM

Product Sales:

- The primary revenue stream will be generated through the sale of the "Better-Fly" automated butterfly rearing system to small-scale butterfly farmers in Marinduque.

Pricing Strategy:

- The system will be priced to ensure profitability while remaining accessible to the target market.
- Consideration will be given to a tiered pricing model or payment plans to accommodate farmers with varying financial capabilities.
- The pricing will factor in the cost of production (40,600 PHP), desired profit margin, and perceived value by the farmers.

  The target is a 50% profit margin:

  - Cost per unit: 40,600 PHP
  - Desired profit per unit: 40,600 PHP * 0.50 = 20,300 PHP
  - Selling price per unit: 40,600 PHP + 20,300 PHP = 60,900 PHP

Sales Target:

- A realistic sales target will be determined within the first year of launch to recoup development costs and generate profit.
- Using the selling price of 60,900 PHP:
- To recover the initial 40,600 PHP cost, at least one unit needs to be sold.
- To generate a meaningful profit, a higher sales volume is needed (e.g., at least 20 systems in the first year

Cost of Revenue:

Direct Costs:

- Production costs of the "Better-Fly" system (40,600 PHP for the first unit).
- Costs of materials for each additional unit.
- Labor costs for assembly

Indirect Costs:

- Marketing and sales expenses.
- Customer support costs.

Activity No. 7 Task Assignment Matrix

No.	Name of Student	Assigned Task	Accomplished Task	Date Accomplished
1	Casimero, Yza S. (Leader)	Revenue Streams	Revenue Streams	March 17, 2025
2	Ponce, Mark Daniel O. (Member)	Cost Structure	Cost Structure	March 17, 2025
3	Saez, Brylle Ildefon V. (Member)	Customers & Customer Relationships	Customers & Customer Relationships	March 17, 2025
4	Sales, Sunshine M. (Member)	Partnership & Channels	Partnership & Channels	March 17, 2025
5	Belarmino, Joanna May M. (Member)	Distribution (Channels)	Distribution (Channels)	March 17, 2025
6	Gonzales, John Dominique N. (Member)	Key Resources	Key Resources	March 17, 2025
7	Moros, Julian Dave C. (Member)	Key Activities	Key Activities	March 17, 2025